

Supplementary for Consumption Theory

Abstract

Consumption theory holds a significant position in the field of macroeconomics, and in this lecture note, we will discuss the four main theories associated with it. To ensure clarity and accessibility, I have omitted the mathematical derivations and instead opted for a simplified explanation using straightforward language. These explanations draw inspiration from renowned macroeconomic textbooks, which are listed below for further reference.

We begin with John Maynard Keynes's General Theory, published in 1936. Keynes made the consumption function central to his theory of economic fluctuations, and it has played a key role in macroeconomic analysis ever since. Let's consider what Keynes thought about the consumption function and then see what puzzles arose when his ideas were confronted with the data.

Absolute Income Hypothesis AIH

Definition 0.1. *Individual consumption is entirely determined by an individual's current absolute income level and is independent of both the relative income level of others and the individual's future income level.*

Keynes believed that people's fundamental psychological law is that consumption increases with income, but the increase in average consumption level is relatively small (APC is small). From this concept, it can be inferred that **consumption is a function of income, and as income increases, the proportion of consumption to disposable income decreases.**

Function

We list several functions used in Keynesian consumption theory below.

Consumption function: describe the relation between consumption and disposable income.

$$C = \bar{C} + cY_d$$

Average propensity to consume, APC : the ratio of consumption to disposable income

$$APC = \frac{C}{Y_d}$$

Marginal propensity to consume, MPC: the amount by which consumption changes when disposable income increases by one dollar. The MPC is between zero and one.

$$MPC = \frac{\partial C}{\partial Y_d} = c$$

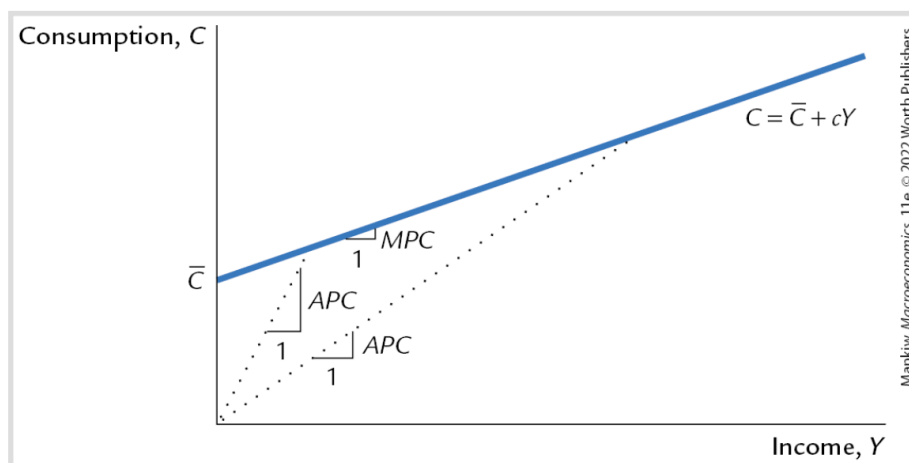


FIGURE 20-1

The Keynesian Consumption Function This figure graphs a consumption function with the three properties that Keynes conjectured. First, the marginal propensity to consume c is between zero and one. **Second, the average propensity to consume falls as income rises.** Third, consumption is determined by current income.

Note: The marginal propensity to consume MPC is the slope of the consumption function. **The average propensity to consume $APC = C/Y$ equals the slope of a line drawn from the origin to a point on the consumption function.**



Figure 1: Keynes consumption function

The Consumption Puzzle

In the 1940s, the economist Simon Kuznets constructed new aggregate data on consumption and income dating back to 1869 — work that later won him a Nobel Prize. Kuznets discovered that the ratio of consumption to income was remarkably stable from decade to decade, despite large increases in income over the period he studied. Again, Keynes's conjecture that the average propensity

to consume would fall as income rose appeared not to hold. The findings of Kuznets indicated that **the average propensity to consume is fairly constant over long periods of time**. This fact presented a puzzle that motivated much of the subsequent research on consumption. Economists wanted to know why some studies confirmed Keynes's conjectures and others refuted them. That is, why did Keynes's conjectures hold up well in studies of household data and in studies of short time-series but fail when long time-series were examined? Figure 2 illustrates the puzzle. The evidence suggested two consumption functions. For **the household data and for the short time-series, the Keynesian consumption function appeared to work well**. But for the **long time-series, the consumption function appeared to exhibit a constant average propensity to consume**. In Figure 2, these two relationships between consumption and income are called the short-run and long-run consumption functions. Economists needed to explain how these two consumption functions could be consistent with each other.

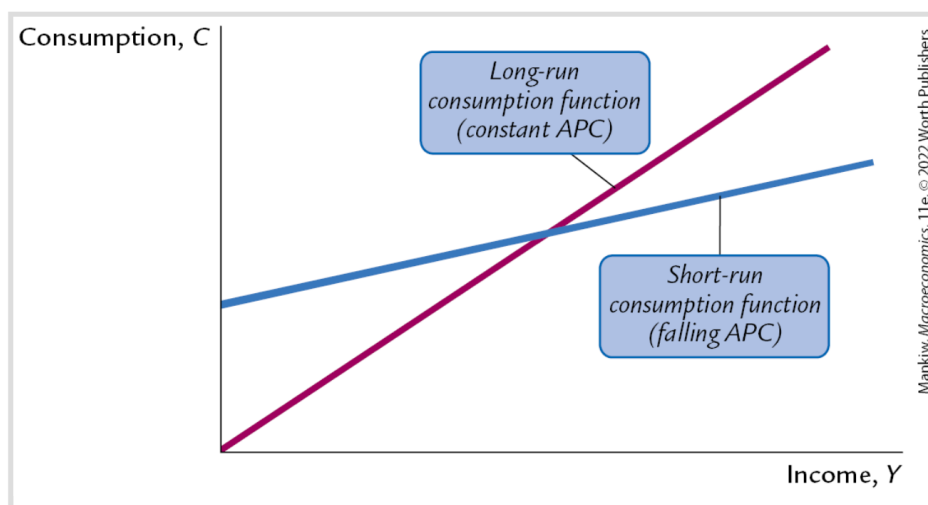


FIGURE 20-2

The Consumption Puzzle Studies of household data and short time-series found a relationship between consumption and income like the one Keynes conjectured. In the figure, this relationship is called the short-run consumption function. But studies of **long time-series found that the average propensity to consume did not vary systematically with income. This relationship is called the long-run consumption function**. Note that the short-run consumption function has a falling average propensity to consume, whereas the long-run consumption function has a constant average propensity to consume.



Figure 2: The Consumption Puzzle

Life-Cycle Hypothesis LCH

Definition 0.2. *The life-cycle hypothesis views individuals, instead, as planning their consumption and savings behavior over long periods with the intention of allocating their consumption in the best possible way over their entire lifetimes.*

Assumption

1. Individuals are rational and their objective is to maximize utility.
2. Individuals have forward-looking behavior, meaning they have perfect foresight to use their lifetime income to plan their consumption and savings. They aim to equalize their lifetime income with consumption. In other words, individuals try to consume the same amount each year.
3. Income is derived from two sources: W represents wealth (non-labor income) and Y represents income (labor income).
4. In the short run, W remains fixed, while in the long run, W can be changed.

Suppose a consumer expects to live another T years, has wealth of W , and expects to earn income Y per year until he retires R years from now. What level of consumption will the consumer choose if he wants stable consumption over the course of his life?

Function

The consumer's lifetime resources are composed of initial wealth W and lifetime earnings. (For simplicity, we assume the interest rate is zero; if the interest rate were greater than zero, we would need to take into account the interest earned on savings.) The consumer can divide up his lifetime resources among his T remaining years of life. To achieve the smoothest possible path of consumption over his lifetime, he divides the total of $W + RY$ equally among the T years and each year consumes

$$C = (W + RY)/T$$

We can write this person's consumption function as

$$C = \frac{W}{T} + \frac{R}{T}Y$$

For example, if the consumer expects to live for 50 more years and work for 30 of them, then $T = 50$ and $R = 30$, so his consumption function is

$$C = 0.02W + 0.6Y$$

This equation says that consumption depends on both income and wealth. An extra \$1 of income per year raises consumption by \$0.60 per year, and an extra \$1 of wealth raises consumption by \$0.02 per year.

If everyone plans consumption like this, the aggregate consumption function is much the same as the individual one: Aggregate consumption depends on both wealth and income. That is, the economy's consumption function is

$$C = \alpha W + \beta Y$$

where the parameter α is the marginal propensity to consume out of wealth, and the parameter β is the marginal propensity to consume out of income.

Implication

This life-cycle model of consumer behavior can solve the consumption puzzle. According to the life-cycle consumption function, the average propensity to consume is

$$\frac{C}{Y} = \alpha\left(\frac{W}{Y}\right) + \beta$$

Because wealth does not vary proportionately with income from person to person or from year to year, we should find that high income corresponds to a low average propensity to consume when we look at data across individuals or over short periods of time. But over long periods of time, wealth and income grow together, resulting in a constant ratio W/Y and thus a constant average propensity to consume. In this way, LCH resolved the consumption puzzle posed by Simon Kuznets's data.

The life-cycle model makes many other predictions as well. Most importantly, it predicts that saving varies over a person's lifetime. If a person begins adulthood with no wealth, he will accumulate wealth during his working years and then run down his wealth during retirement. Figure 3 shows the consumer's income, consumption, and wealth over his adult life. According to the life-cycle hypothesis, because people want to smooth consumption over their lives, the young who are working save, and the old who are retired dissave.

Motivated by this model, many economists have studied consumption and saving among the elderly. They often find that the elderly do not dissave as much as the model predicts. In other words, the elderly do not run down their wealth as quickly as one would expect if they were smoothing their consumption over their remaining years of life. One reason may be the uncertainty they face regarding life span and future medical expenses. Another reason may be that they want to leave bequests to their descendants. Providing for retirement is one motive for saving, but other motives appear to be important as well.

Conclusions

There are several important conclusions about LCH:

1. Lifetime income may vary, but it can be fully anticipated.
2. Pursuing maximum lifetime utility through whole life budget.
3. The magnitude of consumption fluctuations will be smaller than income fluctuations.
4. Consumption smoothing.
5. Consumption is influenced by lifetime income, not by current income.

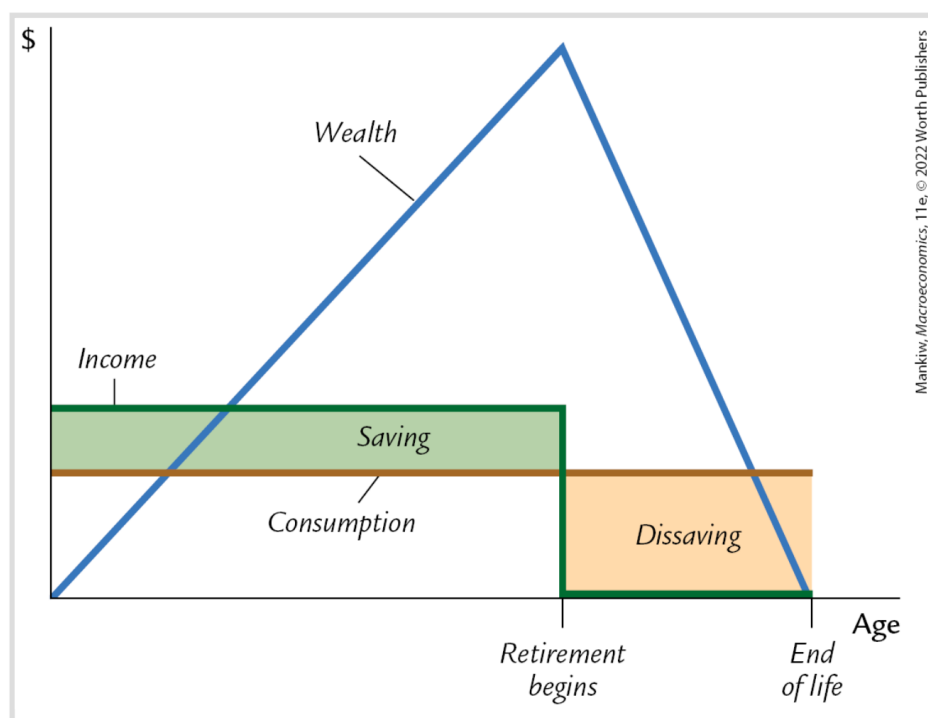


FIGURE 20-5

Consumption, Income, and Wealth over the Life Cycle If the consumer smooths consumption over his life (as indicated by the horizontal consumption line), he will save and accumulate wealth during his working years and then dissave and run down his wealth during retirement.



Figure 3: Consumption, Income, and Wealth over the Life Cycle

Permanent Income Hypothesis PIH

Definition 0.3. *Permanent income is the steady rate of expenditure a person could maintain for the rest of his or her life, given the **present level** of wealth and the income earned now and in the future.*

Friedman's permanent-income hypothesis complements Modigliani's life-cycle hypothesis: Both argue that consumption does not depend on current income alone. But unlike the life-cycle hypothesis, which emphasizes that income follows a regular pattern over a person's lifetime, the permanent-income hypothesis emphasizes that **people experience random changes in income from year to year.**

Assumption

1. **No borrowing constraint and individuals' objective is to maximize utility.**
2. Consumption is the function of permanent income $C = \alpha Y^P$, Friedman thinks that α approximates to 1.
3. **Individuals have forward-looking behavior**
4. Current income Y is the sum of two components: **permanent income Y^P** and **transitory income Y^T** . That is $Y = Y^P + Y^T$. Permanent income is the part of income that people expect to persist into the future. Transitory income is the part of income that people do not expect to persist. **Put differently, permanent income is average income, and transitory income is the random deviation from that average.**

To see how we might separate income into these two parts, consider these examples:

- Nia, who has a law degree, earned more this year than Ethan, who is a high school dropout. Nia's higher income resulted from higher permanent income because her education will continue to provide her a higher salary.
- Lily, a Florida orange grower, earned less than usual this year because a freeze destroyed her crop. Julio, a California orange grower, earned more than usual because the freeze in Florida drove up the price of oranges. Julio's higher income resulted from higher transitory income because he is no more likely than Lily to have good weather next year.

These examples show that different forms of income have different degrees of persistence. **A good education provides a permanently higher income, whereas good weather provides only transitorily higher income.** Although one can imagine intermediate cases, it is useful to keep things simple by supposing that there are only two kinds of income: **permanent and transitory.**

Friedman reasoned that consumption depends primarily on permanent income because consumers use saving and borrowing to smooth consumption in

response to transitory changes in income. For example, if a person received a permanent raise of \$10,000 per year, his consumption would rise by about as much. Yet if a person won \$10,000 in a lottery, he would not consume it all in one year. Instead, he would spread the extra consumption over the rest of his life. If we assume an interest rate of zero and a remaining life span of 50 years, consumption would rise by only \$200 per year in response to the \$10,000 prize. Thus, consumers spend their permanent income, but they save most of their transitory income. Friedman concluded that we should view the consumption function as approximately

$$C = \alpha Y^P$$

where α is a constant that measures the fraction of permanent income consumed. The permanent-income hypothesis, as expressed by this equation, states that consumption is proportional to permanent income.

Implication

The permanent-income hypothesis solves the consumption puzzle by suggesting that the standard Keynesian consumption function uses the wrong variable. While many studies of the consumption function try to relate consumption to current income Y , the permanent-income hypothesis asserts that consumption depends on permanent income. Friedman argued that this *errors-in-variables problem* explains the seemingly contradictory findings.

Let's see what Friedman's hypothesis implies for the average propensity to consume. Divide both sides of his consumption function by Y to obtain

$$APC = \frac{C}{Y} = \alpha \frac{Y^P}{Y}$$

According to the PIH, the average propensity to consume depends on the ratio of permanent income to current income. When current income temporarily rises above permanent income, the average propensity to consume temporarily falls; when current income temporarily falls below permanent income, the average propensity to consume temporarily rises.

Now consider the studies of household data. Friedman reasoned that these data reflect a combination of permanent and transitory income. Households with high permanent income have proportionately higher consumption. If all variation in household income came from the permanent component, the average propensity to consume would be the same in all households. But some of the variation in income comes from the transitory component, and households with high transitory income do not have higher consumption. Therefore, researchers find that high-income households have, on average, lower average propensities to consume.

Similarly, consider the studies of time-series data. Friedman reasoned that year-to-year fluctuations in income are dominated by transitory income. Therefore, years of high income should be years of low average propensities to consume. But over long periods of time — say, from decade to decade — the variation in income comes from the permanent component. Hence, in a long time-series, one should observe a constant average propensity to consume, as Kuznets found.

Conclusion

There are several important conclusions about PIH:

1. Consumption smoothing.
2. Transitory income has small effect(almost zero) on consumption.
3. Permanent income has large effect on consumption.

Reference

1. Macroeconomics, N. Gregory Mankiw, 11 Edition
2. Macroeconomics, Rudiger Dornbushch, Stanley Fischer, Richard Startz, 13 Edition